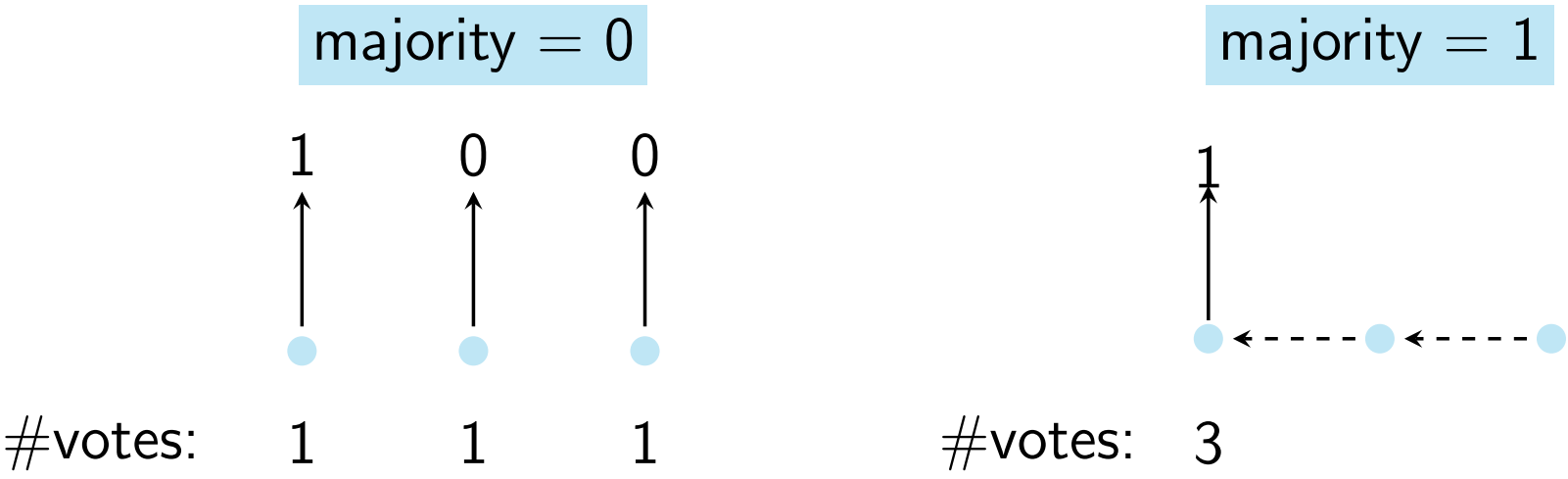
Liquid democracy

Jakub Svoboda
with Michelle Yeo, Seth Gilbert, Stefan Schmid, and Krishnendu Chatterjee



Direct vs liquid democracy



Why liquid democracy?

Practicality

- Motivated by blockchain (DAOs, delegated staking)
- Used in the real world too.

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Practicality

- Motivated by blockchain (DAOs, delegated staking)
- Used in the real world too.

Properties

- Individual: less informed voters can delegate instead of abstaining.
- Societal: decisions may improve when voters delegate to more informed voters.

Better decisions?

- What does “better” mean?
- How is a decision made?
- Is delegation always beneficial?

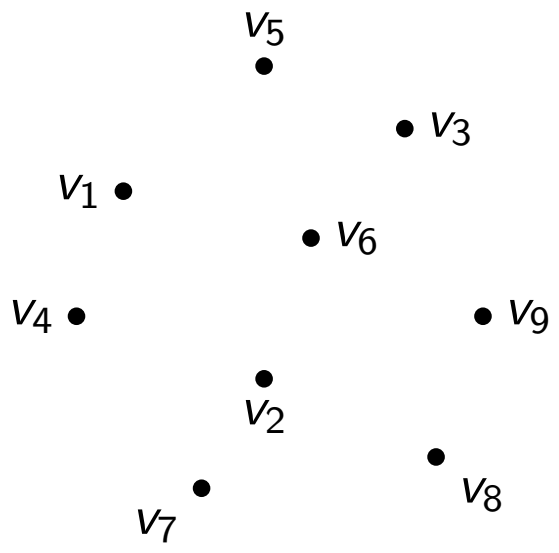
Model and setting

Binary voting with ground truth

Each voter v_i has a competency p_i (probability of voting for the ground truth).

Competencies:

$p_1 = 0.8$
$p_2 = 0.6$
$p_3 = 0.5$
$p_4 = 0.4$
$p_5 = 0.3$
$p_6 = 0.3$
$p_7 = 0.2$
$p_8 = 0.2$
$p_9 = 0.1$




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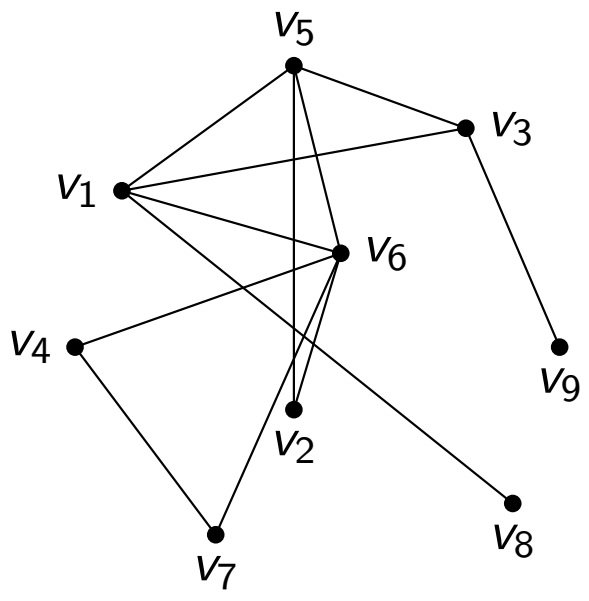
Model and setting

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Each voter v_i has a competency p_i. Each voter is a vertex in a graph.

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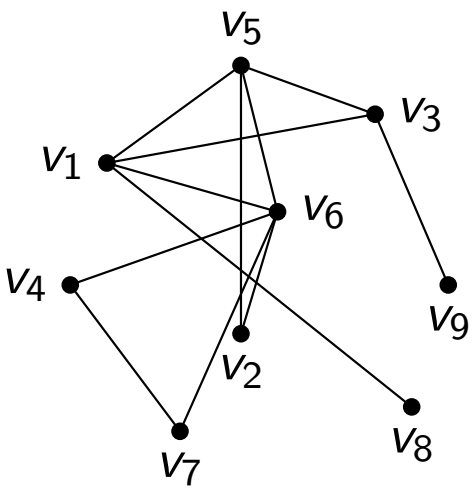
Model and setting

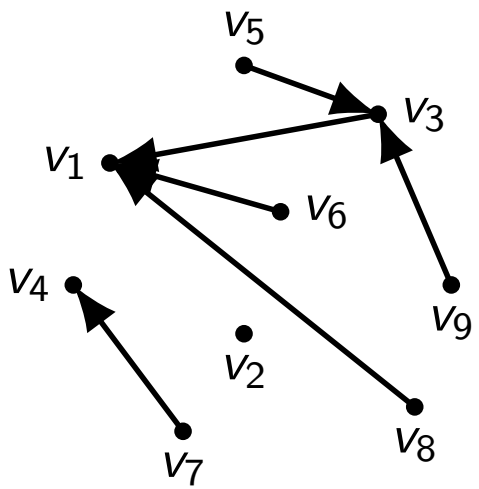
Binary voting with ground truth

Each voter v_i has a competency p_i. Each voter is a vertex in a graph. Each voter can delegate to a neighbor.

Competencies:

$p_1 = 0.8$
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How to think about it

Distributed computing

Every voter

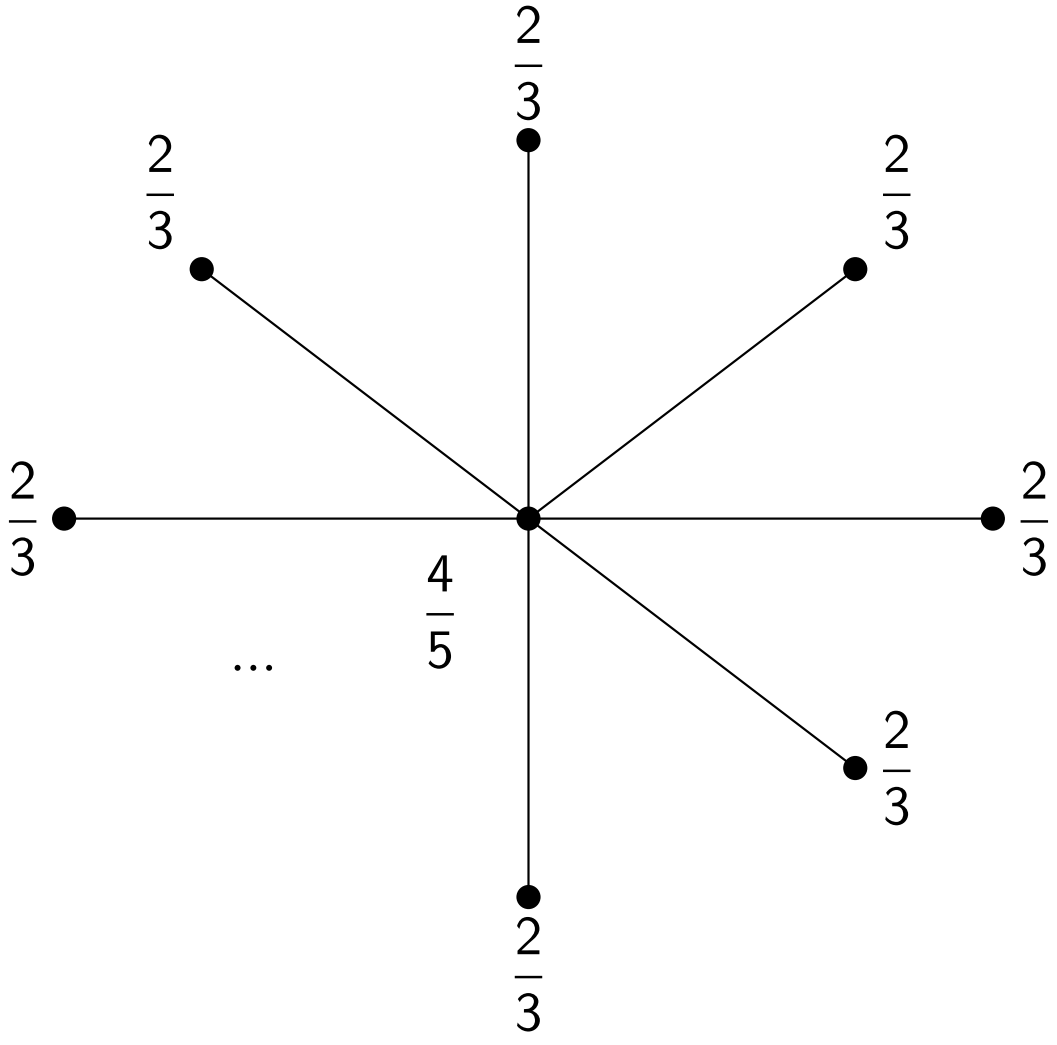
- knows neighbors
- does not see the competencies (even their own)
- has an approval set (competency higher by α)

Question

Is there instance where the delegation is harmful?

Question

Is there instance where the delegation is harmful?



Aims

- **Positive gain:** there exist instances where delegation performs better.
- **Do no harm:** for all instances, the loss asymptotically goes to 0 as the graph grows.
- **Strong positive gain:** for all instances, delegation performs better.

Why is this hard

1. Delegation creates **positive correlations** between random votes.
2. To get "do no harm", one needs **sufficient variance** in the voting outcome.

Technical obstacle

Standard concentration tools are much better behaved for independence or negative correlation than for positive correlation.

Ideas for solution

Delegation rule

For each voter v_i, let $J(i)$ be the approval set. If $|J(i)| > t$, then v_i delegates to a random voter in $J(i)$.

Intuition

Sufficient variance in the voting outcome helps ensure the *do no harm* property.

Results

Theorem (informal)

There are local delegation mechanisms such that strong positive gain and do no harm are satisfied on complete and random d-regular graphs.

Theorem (informal)

There are local delegation mechanisms such that strong positive gain and do no harm are satisfied on bounded-degree and bounded-minimal-degree graphs.

Implications

- Concentration of influence can cause harm.
- Structural asymmetry in the graph matters.
- Best case: complete graph, where everyone knows and trusts everyone.

No more ground truth

New setting

Now voters have preferences in $\{0, 1\}$ rather than a common ground truth.

- Competencies measure how likely voters are to vote for their preferred outcome.
- There may be indecision or stochasticity between preference and cast vote.
- A faction wants to design delegation to maximize the probability of winning.

Strategies

Possible strategies include *Dictator*, *Direct*, and *Representative(k)*.

Complexity

Theorem

Finding optimal strategies in the general setting is PSPACE-complete.


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Complete graph

Simple setting

- On complete graph (everyone can delegate to everyone)
- Everyone with the same preference has the same competence. c_0 and c_1
- Preference 0 has proportion p (and opponent $1 - p$)

Equal proportion

Question

What is optimal strategy for equal faction sizes: $p = 0.5$.

Equal proportion

Question

What is optimal strategy for equal faction sizes: $p = 0.5$.

Theorem (informal)

If both factions play strategically, then both playing Dictator is the unique Nash equilibrium.

Faction 0 is stronger

Condition

$c_1 = 1$ and $p \cdot c_0 \geq 0.5$.

Theorem (informal)

Faction 0 *wins over faction* 1 *by playing Direct.*

Faction 0 is weaker

Condition

$c_1 = 1$ and $p \cdot c_0 < 0.5$.

Theorem (informal)

Delegation improves the winning probability over direct voting.

Implications

- A weaker party should still win sometimes.
- Direct voting can be unfair.
- Delegation can improve fairness for a weaker but larger party.
- Delegation compensates for lower competency by size.

Last slide

Thank you for your attention!